

Quick Rad AI - On-Prem DICOM Forwarder

Deployment Package Guide for Hospital IT / Information Security

Vendor: GenZ AI Labs

This document explains, in plain terms, every file in the download package and exactly what it does - so your IT and security teams can review it before installation. The full source code is included in the package and is open for inspection; nothing is hidden, compiled, or obfuscated.

1. What this software does

A small store-and-forward gateway that runs on ONE hospital PC that has internet access. Imaging scanners / PACS on the hospital network send DICOM studies to this PC (a standard DICOM C-STORE). The service receives them and forwards each study to the Quick Rad AI cloud for analysis. If the internet or cloud is temporarily down, studies are kept on disk and retried until they are delivered - nothing is lost. It is a DICOM relay only: it does not read, modify, or scan anything else on the PC or network.

2. Package (folder) structure

After unzipping the download, the folder contains these files:

File	Type	What it is
qr_forwarder.py	Program	The forwarder itself (readable Python source).
forwarder_config.json	Settings	Plain-text configuration (ports, names, endpoint).
install.bat	Installer	One-click install script (right-click, Run as admin).
run.bat	Test tool	Optional: run once in a console to watch live logs.
requirements.txt	Manifest	Lists the two open-source libraries used.
README.txt	Docs	Step-by-step install and management instructions.
qr_forwarder_data\	Runtime	Created on first run: holds studies + the log file.

3. What each file does (in detail)

qr_forwarder.py (the program)

- Receives DICOM images from the scanner on the local network (port 11120).
- Groups images into complete studies, then forwards each study to the cloud.
- Store-and-forward with retry: keeps studies on disk and retries every ~30s if the cloud/internet is down, so no study is ever lost.
- Writes a plain-text activity log (forwarder.log).
- Automatically deletes already-sent studies after a few days to save space.
- Full source is included and readable - no compiled or hidden code.

forwarder_config.json (settings)

- A plain-text file you can open in Notepad.
- Sets the branch/site name (DICOM AE title), the local and cloud ports, the cloud endpoint, and how long sent studies are kept.
- Configuring a new site is a one-line change - no code editing.

install.bat (installer)

- Run once by right-clicking and choosing 'Run as administrator'.
- Installs the two open-source libraries (pynetdicom, pydicom) via pip.
- Opens the Windows Firewall for inbound TCP 11120 so the scanner can reach this PC.
- Registers an auto-start job so the forwarder keeps running.
- Fully reversible - the exact removal command is printed at the end.

run.bat (test tool)

- Optional. Starts the forwarder in a visible console window to watch live logs during testing. Not used for normal 24/7 operation.

requirements.txt (dependency list)

- Names the only two third-party libraries used: pynetdicom and pydicom - the standard, widely-used open-source DICOM libraries. install.bat installs them automatically.

README.txt (instructions)

- Human-readable install steps and the commands to check, stop, or remove the job.

4. Network behavior (for firewall / security review)

Direction	Host	Port	Purpose
Inbound (LAN only)	this PC	TCP 11120	receive studies from the scanner
Outbound (internet)	quickradai.genzailabs.com	TCP 11112	forward studies to the cloud

Listens on the hospital LAN only - no inbound connections from the internet. One outbound destination, a named host that can be allowlisted. No telemetry and no remote-control channel.

5. What it does NOT do

- No hidden or obfuscated code - the complete source is in this package.
- Handles DICOM imaging only - it does not read, modify, or scan other files or drives.
- No remote-access / backdoor capability; it connects to exactly one cloud destination.
- No third-party analytics or telemetry.
- Imaging data is stored on-premise and auto-purged after the configured number of days.

6. How to check, stop, or remove it

```
schtasks /query /tn "QuickRadForwarder" (check status)
schtasks /end /tn "QuickRadForwarder" (stop)
schtasks /delete /tn "QuickRadForwarder" /f (remove completely)
```

Activity log: qr_forwarder_data\forwarder.log

GenZ AI Labs - Quick Rad AI - Cloud endpoint: quickradai.genzailabs.com:11112 (AE QUICKRAD_AI). For questions, contact your GenZ AI Labs representative.